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Electronic conductivity in the SiO_2–PbO–Fe_2O_3 glass containing magnetic nanostructures

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ABSTRACT

The linear impedance spectra of iron–silicate–lead glass samples were measured in the frequency range from 1 MHz to 1 MHz and in the temperature range from 153 K to 423 K. The structure was investigated by means of XRD and atomic force microscopy. Local electrical and magnetic properties of the samples were tested with the aid of electrostatic force microscopy (EFM) and magnetic force microscopy (MFM).
The obtained results show that the impedance spectra of all samples exhibit one large relaxation process and some additional small relaxation is present in glass containing 20 mol% and 25 mol% Fe_2O_3. The AFM micrographs show that in a subset of samples some nanostructures exist. Their size, shape, and quantity depend on the amount of Fe_2O_3 in the composition of the glass. The samples containing an amount of 15 mol% of Fe_2O_3 include the biggest nanostructures which are ball-shaped. EFM and MFM micrographs show that nanostructures exhibit different electrical and magnetic behaviors than the rest of the glass matrix.
On the basis of Jonscher universal dielectric response the temperature dependence of conductivity exponent s was determined and compared to theoretical models proposed by Elliott. It was found that while in the glass containing less than 15 mol% iron oxide, there is only one process responsible for conduction mechanism, the overlap polaron tunneling. In the other samples, a different conduction mechanism may coexist: quantum mechanical tunneling between semiconducting granules.

1. Introduction

Many oxide glass families containing a large amount of transition metal oxides are electronic conducting semiconductors. Their electrical properties are determined by the transition metal ions present in two different valence states [1–3]. Their conductivity was described by a Mott model of small polaron hopping between such ions [4]. Glasses containing iron oxides have attracted the attention of researchers for three decades. Then there were some different models proposed to describe the structure and the conduction mechanism in these glasses [1–7]. Due to the development of new experimental techniques now it has become possible to check the validity of these models.

The investigations of iron–silicate–lead and iron–borate–lead glasses carried out by MacCrone et. al. [5,8–11] suggested that in glass more than 10 mol% Fe_2O_3, iron ions are not randomly distributed. The structural study of these glass systems has found some magnetic inhomogeneities. A model was proposed which suggests that the majority of Fe ions are located in separate clusters which are well ordered and contain various numbers of the iron ions. Ions outside the clusters are coupled in groups of two, three, etc. and form antiferromagnetic alignment. The nearest neighbor inter-ion distance and relative orientation are determined by the covalent bonding with oxygen. Fe ions within clusters form structures very similar to that in the crystalline oxides. Iron oxide exists in three crystalline compounds: FeO, Fe_3O_4, and Fe_2O_3. Each of them has a different structure [3]. The average cluster sizes are about a few nanometers and depend on the iron concentration in the glass and heat treatment. Anderson and MacCrone and Murawski have also suggested that charge transfer in these glasses takes place along the chains of some of the clusters from one electrode to the other. They supposed that not all clusters form the dc conductive path but the number of well joined clusters increases with the increase in the content of Fe ions in the glass. This should explain small differences in dc and ac activation energies and high decrease in dc activation energy with an increase of amount of Fe_2O_3 in glass [3,5].

The other model describing silicate–lead and borate–lead glass systems containing iron ions was proposed by Mendiratta [7]. He found the presence of the α-Fe_2O_3 crystalline phase in glasses containing a higher concentration of Fe ions (0.5 and 0.6 mol). He also showed that conductivity in these glasses exhibit two different activation energies and suggested that lower activation energy is due to conduction through small semiconducting, crystalline microregions, distributed randomly in the amorphous matrix whereas the higher activation energy is caused by hopping over the sites within amorphous regions.

Both cited models suggest that the structure of glass from this system is not homogenous and therefore the conduction is a complex process. The aim of the present study was to investigate the structure

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of glasses of this family by means of advanced microscopy and to determine its influence on their electrical properties.

2. Experimental

Two different sets of glass samples were prepared. The first group of samples has the composition of $(50 - 0.5x)SiO_2$–$(50 - 0.5x)PbO$–xFe_2O_3, where x = 2, 5, 10, 15, 20 and 25 (in mol%). Composition of the second group was $50SiO_2$–$(50 - x)PbO$–xFe_2O_3, where x = 15, 20 and 25 (in mol%). The compositions of the studied glasses are listed in Table 1. All samples were prepared by the conventional melt quenching technique; the melting was conducted in air at 1623 K in alumina crucibles. The melts were poured on a brass plate preheated to 573 K and formed into button-shaped samples. Afterwards pellets were annealed at 673 K, cooled with the furnace and polished to the thickness of about 1 mm.

For the electrical conductivity measurements, golden electrodes were evaporated at both sides of preheated samples in vacuum. The linear impedance spectra were measured in the frequency range of 1 MHz to 1 MHz and in the temperature range of 153 K to 423 K by Novocontrol Concept 40 broadband dielectric spectrometer. The microstructure of all samples was determined by means of atomic force microscopy at their fresh fractures with the use of NT-MDT Ntegra AFM. Magnetic properties were analyzed by magnetic force microscopy (MFM). The presence of crystalline phases was examined by the X-ray diffraction method with the use of a Philips X'Pert Pro MPD system with CuKα radiation.

3. Results

Fig. 1 displays imaginary and real parts of impedance spectrum versus frequency of the $42.5SiO$–$42.5PbO$–$15Fe_2O_3$ glass. The behavior of both impedance parts suggests one relaxation process. The inset shows the plot of imaginary versus real part of impedance (Nyquist plot). It presents a single semicircle. All samples show similar impedance spectra and Nyquist plots in the whole temperature range.

Fig. 2a presents the frequency dependence of the real part of conductivity measured at different temperatures for the $15Fe_2O_3$–$50SiO_2$–$35PbO$ glass sample. The plot consists of two regions: dc plateau dominating in the low frequency region and ac component increasing with frequency in a linear fashion (power law behavior). Fig. 2b shows frequency dependence of real part of conductivity measured at various temperatures for the $37.5SiO_2$–$37.5PbO$–$25Fe_2O_3$ glass. It may be noticed that a small relaxation process appears in the high frequency domain which shifts with the increase in temperature. Similar behavior of conductivity occurs in the glass containing 20 mol% Fe_2O_3.

Fig. 3 shows the atomic force microscopy fracture images of $(50 - 0.5x)SiO_2$–$(50 - 0.5x)PbO$–xFe_2O_3 glass for x = 10, 15 and 25. As can be seen from Fig. 3a the glass containing 10% of Fe_2O_3 is homogenous. Glass samples with less than 10% of iron oxide show a similar structure as shown in Fig. 3a. Glass with 15% of iron oxide (Fig. 3b) includes sphere-shaped granules in which the diameter is less than 1 μm.

Table 1
Conductivity parameters.

Composition [mol%]			$\sigma'_{DC\,(T=425\,K)}$ [S/cm]	W [eV]	Structure
Fe_2O_3	SiO_2	PbO			
2	49	49	$1.39 * 10^{-12}$	0.71	Amorphous
5	47.5	47.5	$2.63 * 10^{-10}$	0.66	Amorphous
10	45	45	$1.32 * 10^{-8}$	0.6	Amorphous
15	42.5	42.5	$1.12 * 10^{-7}$	0.56	Nanogranules
20	40	40	$1.02 * 10^{-8}$	0.58	Nanocrystallites
25	37.5	37.5	$2.68 * 10^{-10}$	0.2	Nanocrystallites
15	50	35	$3.78 * 10^{-8}$	0.56	Nanogranules
20	50	30	$1.10 * 10^{-6}$	0.16	Nanocrystallites
25	50	25	$4.9 * 10^{-7}$	0.13	Nanocrystallites

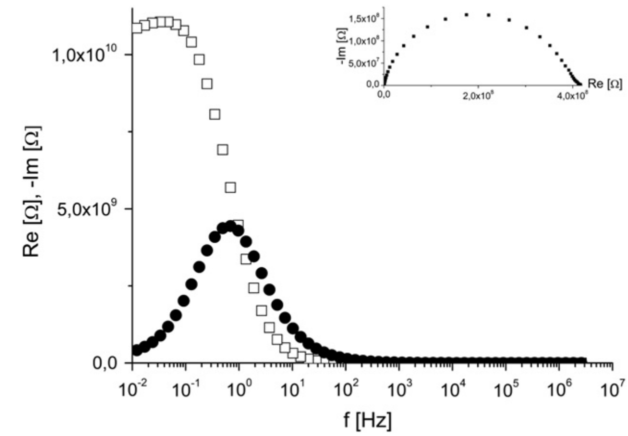

Fig. 1. Imaginary and real parts of impedance spectrum versus frequency of $42.5SiO_2$–$42.5PbO$–$15Fe_2O_3$ glass measured at a temperature of 273 K. Inset shows the Nyquist plot.

Fig. 3c presents the structure of glass containing 20% Fe_2O_3 which contains fairly uniformly ordered oblong-shaped structures. Their length varies between about 100 nm and 350 nm, while average width is about 80 nm.

Fig. 4 presents AFM images for $50SiO_2$–$(50 - x)PbO$–xFe_2O_3 glass fractures for x = 15 and 25 (in mol%). Obtained results present that these glass samples also contain some nanostructures with different sizes (ranges from 40 nm to 400 nm), shape and quantity. In comparison with $(50 - 0.5x)SiO_2$–$(50 - 0.5x)PbO$–xFe_2O_3 glass family, the shape of nanostructures in $50SiO_2$–$(50 - x)PbO$–xFe_2O_3 glass system is more

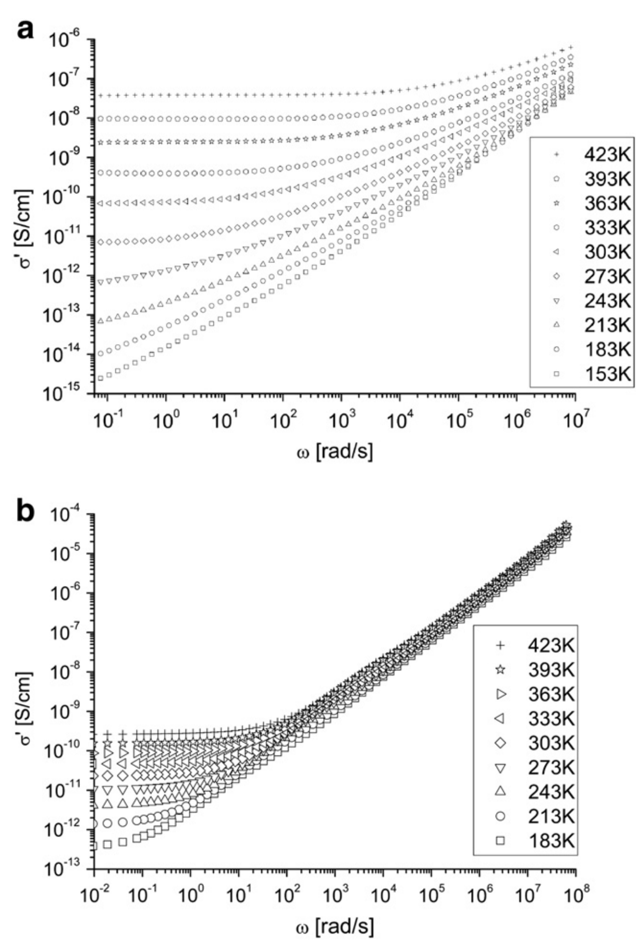

Fig. 2. Real part of conductivity versus frequency measured at different temperatures for (a) $15Fe_2O_3$–$50SiO_2$–$35PbO$ and (b) $37.5SiO_2$–$37.5PbO$–$25Fe_2O_3$ glass samples.

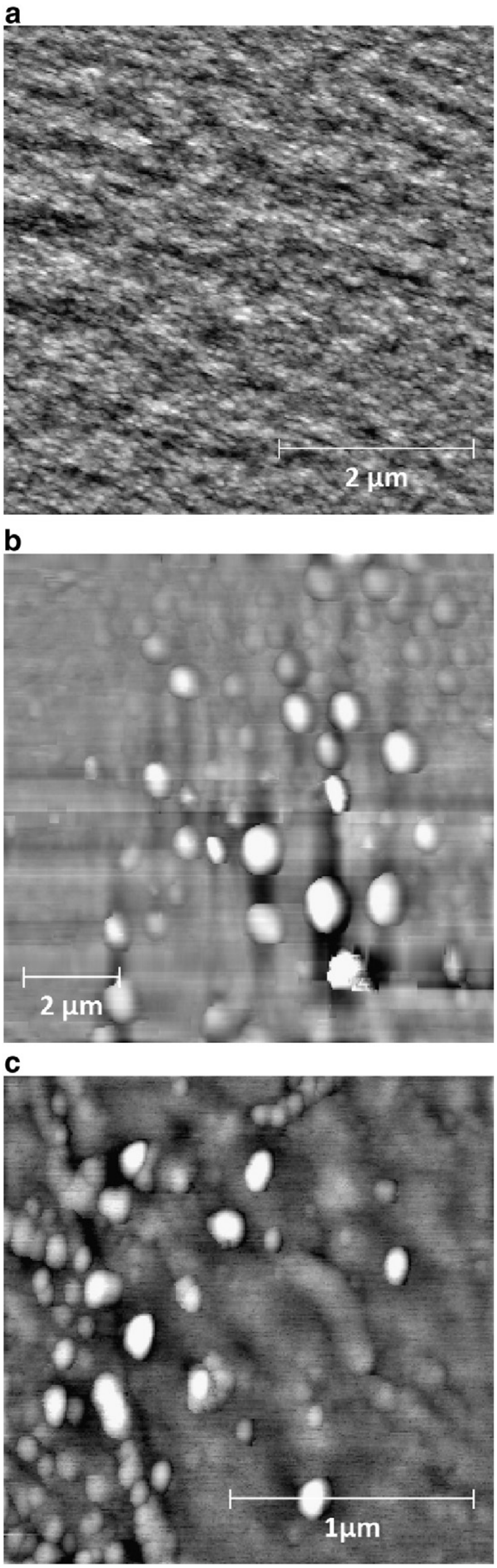

Fig. 3. Atomic force microscopy images of (50 − 0.5x)SiO_2–(50 − 0.5x)PbO–xFe_2O_3 glass fractures in which the content of Fe_2O_3 is respectively: (left) 10, (center) 15 and (right) 25 (in mol%).

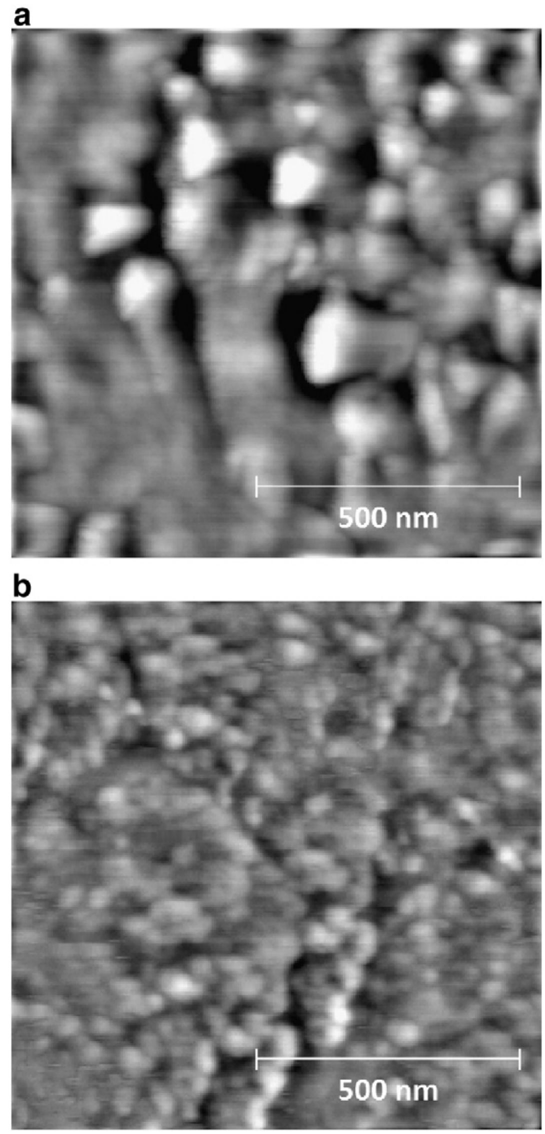

Fig. 4. AFM micrographs of 50SiO_2–(50 − x)PbO–xFe_2O_3 glass fractures in which the content of Fe_2O_3 is respectively: (left) 15 and (right) 25 (in mol%).

spherical. The comparison of obtained pictures shows that size, shape, orientation and quantity of nanostructures present in the structure of iron–silicate–lead glasses depend on the amount of iron oxide and the ratio of silica to the lead oxide. Glass samples containing 15% of iron oxide have the lowest number of nanogranules but they are the largest. The quantity and order of nanostructures increase and their size decreases with an increase in iron oxide contents. The size of granules decreases also with an increase in PbO contents.

X-ray diffraction patterns of 50SiO_2–(50 − x)PbO–xFe_2O_2 and (50 − 0.5x) SiO_2–(50 − 0.5x)PbO–xFe_2O_3 glasses are presented in Figs. 5 and 6. Fig. 5 shows that the XRD patterns for the glass containing 25% of iron oxide consists of amorphous halo and a few reflexes characteristic of crystalline phases. These small peaks correspond to the XRD spectra of Fe_3O_4, Fe_2O_3 and FeO. Fig. 6 shows that the reflexes corresponding to crystalline iron oxides in glasses with 20% of iron oxide are significantly smaller than these in the glasses with 25% of iron oxide. XRD spectra for samples with less than 15% of iron oxide show no peaks and suggest an amorphous structure. It should be noted that despite the fact that the glass with 15% of iron oxide contains nanogranules its XRD spectra show no peaks indicating crystalline phases.

Electrostatic force microscopy image of 42.5SiO_2–42.5PbO–15Fe_2O_3 glass is shown in Fig. 7. Fig. 7a refers to the topography image for the

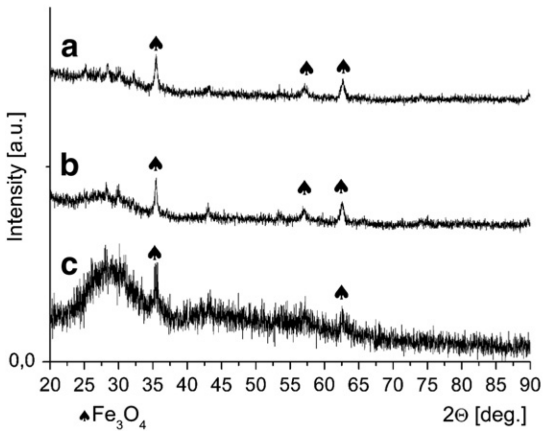

Fig. 5. XRD patterns for: (a) $50SiO_2$–25PbO–$25Fe_2O_3$, (b) $50SiO_2$–30PbO–$20Fe_2O_3$ and (c) $50SiO_2$–35PbO–$15Fe_2O_3$ glass (in mol%).

sample while Fig. 7b corresponds to its EFM image, with the voltage of +4 V. It can be seen that nanostructures have significantly larger electric conductivity than the rest of the glass matrix. Similar images were obtained for glass containing more than 15% Fe_2O_3 and negative voltage applied.

Magnetic force microscopy image of $37.5SiO_2$–37.5PbO–$25Fe_2O_3$ glass is shown in Fig. 8. Fig. 8a refers to the topography image for the sample while Fig. 8b corresponds to its MFM image. It can be seen that the nanogranules exhibit different magnetic properties than the glass matrix. Nanogranules show significantly larger magnetic permeability — they are ferromagnetic. Similar results are obtained for all glasses containing more than 15% of Fe_2O_3.

4. Discussion

The Nyquist plots (Fig. 1) show that behavior of impedance in all glass samples presents one large relaxation process which correspond to the main conduction mechanism. The dependence of the ac conductivity on angular frequency (in amorphous materials) is usually found to obey the form (so called universal dielectric response) [12]:

$$\sigma(\omega) = \sigma_{DC} + A\omega^{s(T)}$$

where σ_{DC} is the dc conductivity and $\sigma_{AC} = A\omega^{s(T)}$ is the ac component. The first term (dc conductivity) dominates in the low frequency region and the second one (power law behavior) in the high frequency region. Fig. 2a shows that the frequency dependence of conductivity for glass samples containing less than 20% of Fe_2O_3 is in good agreement with the Jonscher law. On the other hand glasses with 20% and 25% of iron oxide are not amorphous and do not fulfill this relation (Fig. 2b).

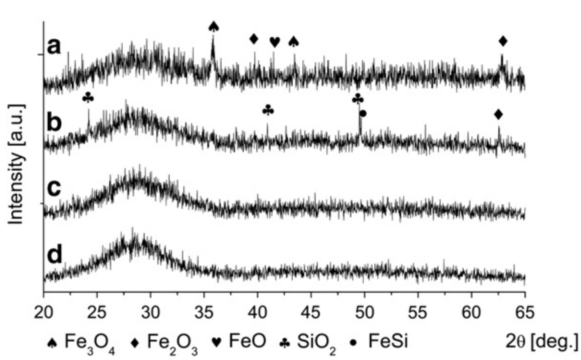

Fig. 6. XRD patterns for: (a) $37.5SiO_2$–37.5PbO–$25Fe_2O_3$, (b) $40SiO_2$–40PbO–$20Fe_2O_3$, (c) $42.5SiO_2$–$42.5SiO_2$–$15Fe_2O_3$, and (d) $42.5SiO_2$–45PbO–$10Fe_2O_3$ glass (in mol%).

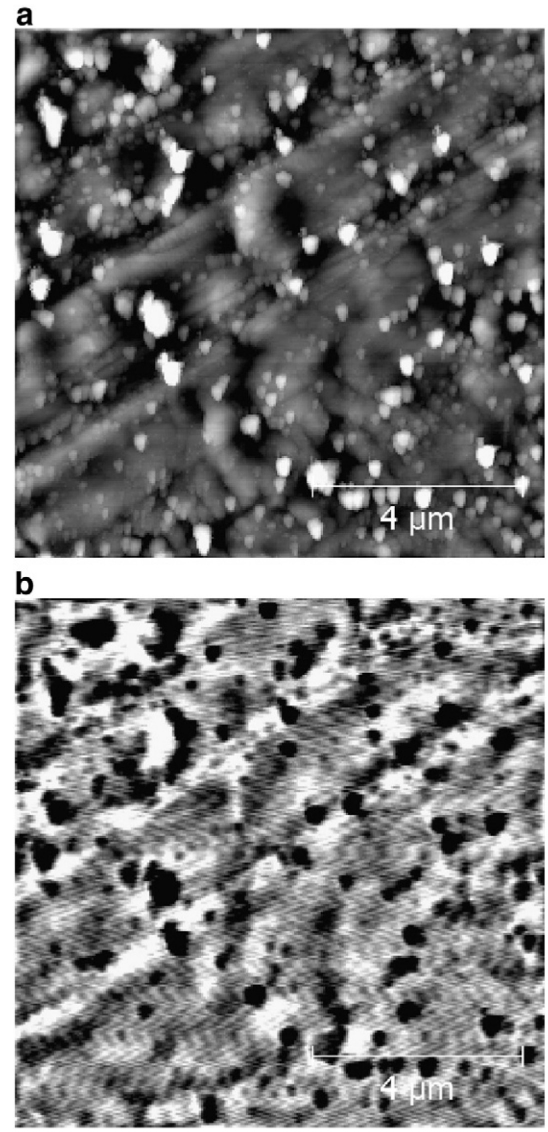

Fig. 7. EFM micrograph of 42.5 SiO_2–42.5PbO–$15Fe_2O_3$ glass (left) morphology, (right) with applied voltage of +4 V.

AFM micrographs (Figs. 3 and 4) and XRD spectra (Figs. 5 and 6) show that samples with different amounts of Fe_2O_3, can be either amorphous and homogenous (less than 15%), or amorphous with spherical nanogranules (15%) or may contain nanocrystallites (more than 15%). We believe that the majority of iron ions in glasses containing 15% and more of Fe_2O_3 are located in nanostructures. For glasses with 20% and 25% of Fe_2O_3 nanocrystallites are formed by iron oxides: FeO, Fe_2O_3 and Fe_3O_4 and in one case of crystalline SiO_2. Micrographs obtained by electrostatic force microscopy (Fig. 7) and magnetic force microscopy (Fig. 8) show that the nanogranules distributed in the glass structure are ferromagnetic and are better conductors than the rest of the glass matrix. It should be noted that ferromagnetic properties and increase of conductivity are shown also by spherical structures in the glass, where XRD diffraction shows no traces of the crystalline phase.

Elliott [13] compared the dependence of exponent s from the universal dielectric response equation on temperature due to various conduction mechanisms in glass. It may be compared to values calculated for our experimental data by means of non-linear data fitting. Fig. 9 shows fitted exponent versus temperature. As can be seen from the figure, for glass $(50 - 0.5x)SiO_2$–$(50 - 0.5x)PbO$–xFe_2O_3 where $x = 5$, 10 and 15 it decreases with the increase in temperature from

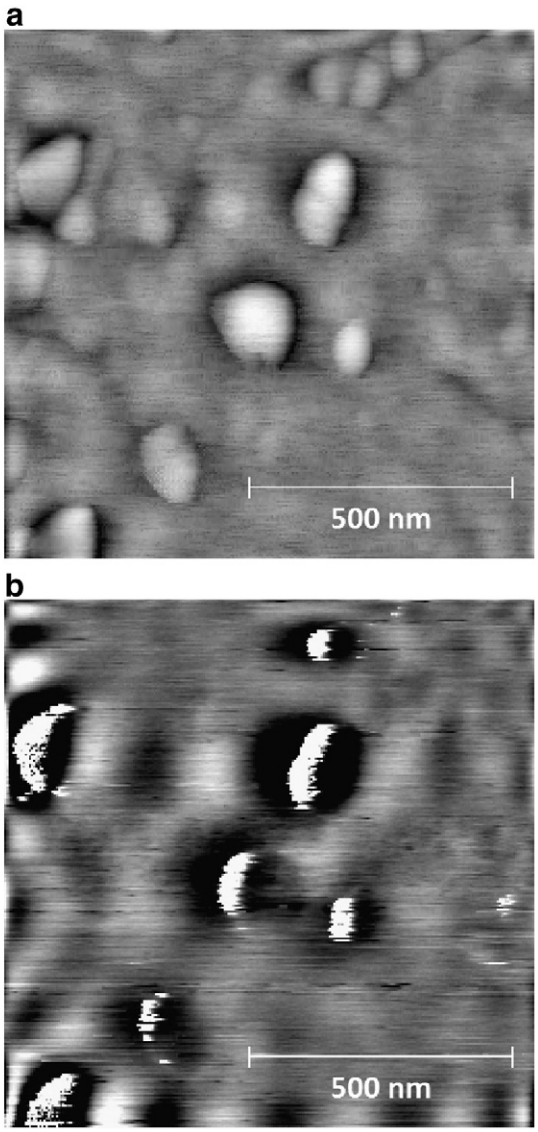

Fig. 8. (Left) morphology and (right) MFM micrograph of $37.5SiO_2$–$37.5PbO$–$25Fe_2O_3$ glass.

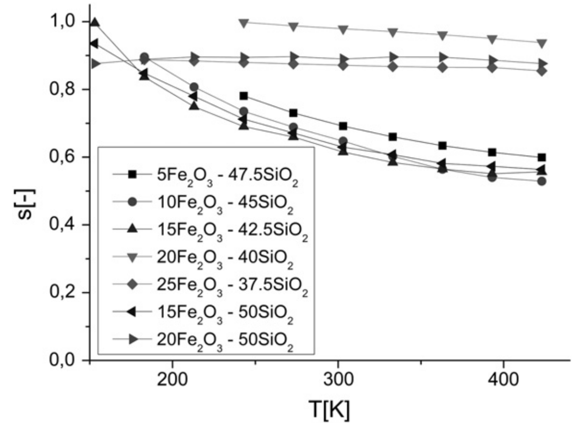

Fig. 9. Temperature dependence of the frequency exponent *s*.

about 1 to about 0.5. In two other cases the exponent *s* is almost constant and its average value is more than 0.9. Similar situation is present in glass $50SiO_2$–$(50 - x)PbO$–xFe_2O_3. Comparing the values of *s* and its dependence on temperature to the models collected by Elliott, we may suppose that in the analyzed glass the conduction process may be due to a couple of mechanisms. In amorphous homogenous iron–silicate glass, the conduction mechanism is the overlap polaron tunneling. In the glasses containing a large amount of iron oxide almost constant and close to one value of *s* the nearly constant loss behavior or more likely quantum mechanical tunneling between semiconducting granules of iron oxides can be proposed.

Table 1 presents dc conductivity measured at 423 K and dc conductivity activation energy for all samples. The values of activation energy decrease with increase in iron oxide amount. For glass samples containing 20% and 25% of Fe_2O_3 the activation energy is quite small which may suggest appearance of other mechanisms of conduction. The $40SiO_2$–$40PbO$–$20Fe_2O_3$ glass sample conductivity has significantly larger activation energy than in the other glasses containing 20% and 25 mol% Fe_2O_3. This glass contains nanocrystallites which are mainly formed from SiO_2. In the other samples nanocrystallites are formed from iron oxides which contributes to the conduction process.

The obtained results are partially in agreement with MacCrone and Anderson [5] and Mendiratta [7] models. We found that in silicate–lead glasses containing more than 10 mol% iron oxide not clusters but nanostructures are formed. The major part of iron ions is located in them and they are separated one from another. The small polaron hopping is not the only conduction mechanism in these glasses, but the conduction process is more complex. MacCrone et. al. concluded that inhomogeneities show magnetic properties what we have also found. However our results do not allow us to determine whether the rest of the matrix is antiferromagnetic.

5. Conclusions

The study of structure of iron–silicate–lead glass shows that some nanogranules are presented in glass with 15% iron oxide. They exhibit ferromagnetic properties and are not visible in XRD spectra. Samples containing more than 15 mol% Fe_2O_3 include a lot of ordered nanostructures. The quantity, size, shape and orientation depend on the composition of glass: amount of Fe_2O_3 and the ratio of SiO_2 to PbO.

Impedance spectra show one large relaxation process which may correspond to polaronic conductivity. However there are some small additional relaxation processes detected in conductivity plots and dielectric permittivity plots for glass samples containing 20 mol% and 25 mol% Fe_2O_3. Activation energies for dc conductivity for these glass samples are surprisingly small. Based on these results and on temperature dependence of exponent *s* we believe that two conduction mechanisms may occur in this glass system: overlap polaron tunneling and quantum mechanical tunneling between semiconducting granules.

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